# ROS 호환 소형 전기 저항 단층촬영 계측보드 개발

# Development of ROS compatible compact PCB for ERT

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## Research Motivation

**Current Challenges**

- Tactile sensing is essential for physical interaction in unstructured environments
- ERT-based large-area robotic skin offers various advantages
- Current measurement equipment is too large for practical implementation

**Technical Needs**

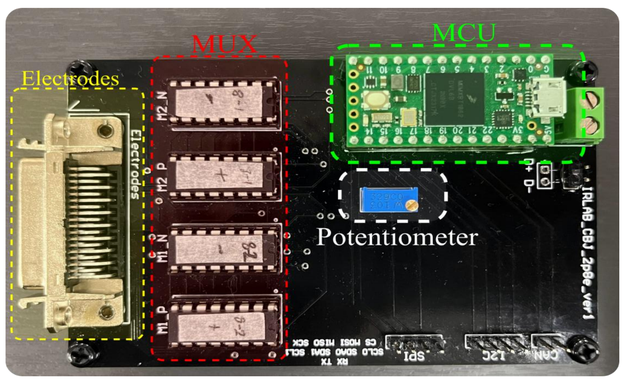
The development of an improves ERT system requires :

- Miniaturization of measurement equipment
- Enhanced software integration capabilities
- Real-time measurement system
- ROS compatibility for practical implementation

**Research Objectives**

This study aims to :

- Develop compact ERT board using small MCU
- Implement µROS for efficient integration
- Achieve high software compatibility
- Enable practical robotic system integration




[Fig. 1] The developed ERT operating board

## ERT Measurement Sensor

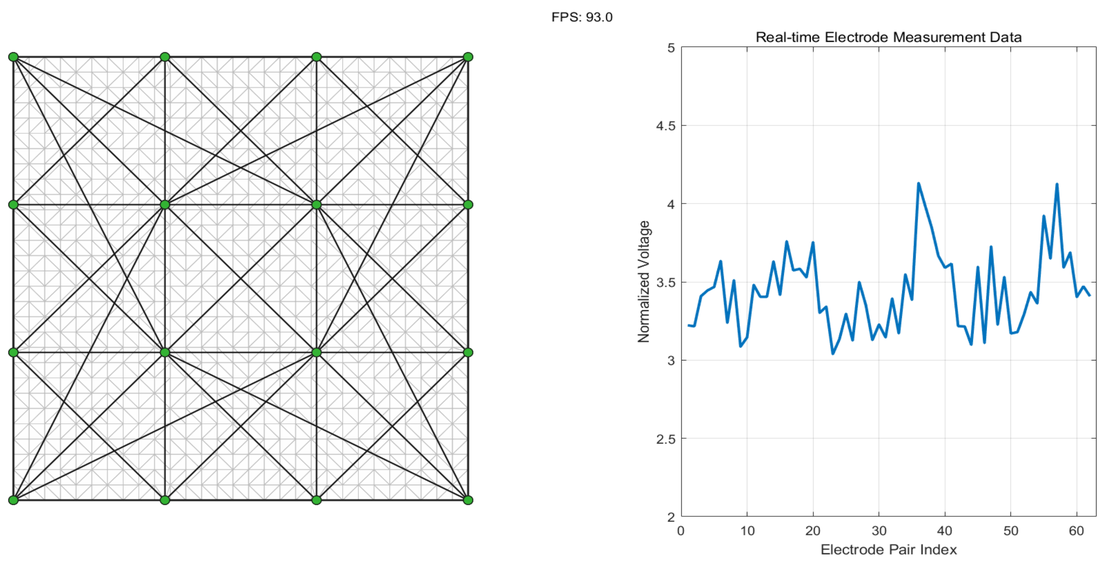
**The sensor board features the following specifications**

- 16 electrodes with 2-pole measurement scheme
- 4 sets of 8-channel multiplexers
- Teensy 4.0 MCU integration
- Compact size : 100*65 [$mm^2$]
- Impedance level control through potentiometer

## Measurement Results

**The measurement system implements**

- Optimized measurement pattern requiring only 62 measurements for 16 electrodes
- Base pattern measurement for reference voltage readings
- Real-time noise filtering using 10-bit ADC with sequential measurement comparison
- Image reconstruction capability through ERT inverse problem solving




[Fig. 2] (a) Base pattern pair; (b) Base pattern measure result

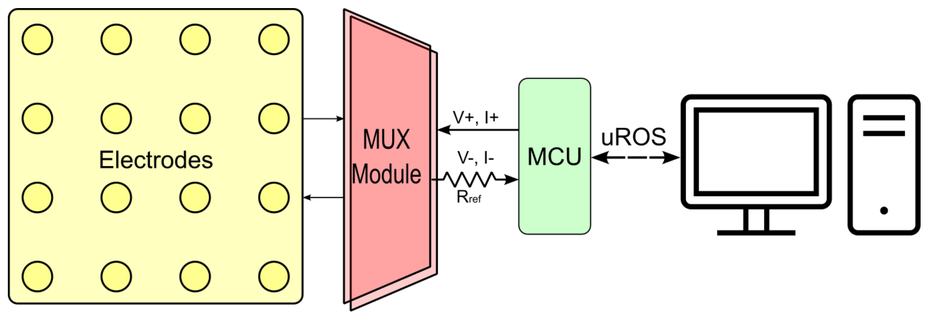
## ROS System Architecture

**The system operates on**

- Ubuntu 22.04 Linux environment
- ROS2 Humble framework
- µROS integration for MCU communication

**Key advantages include**

- Compatibility with various robotic platforms through standard ROS communication protocols
- Distributed computing architecture for efficient data processing
- Easy integration of additional functional nodes




[Fig. 3] ROS system Architecture

## Conclusion

**The developed ERT measurement board achieves**

- Significant size reduction enabling internal robot placement
- Seamless integration with actual robotic systems
- Enhanced usability through ROS compatibility
- Expected contribution to physical human-robot interaction research through improved tactile sensing capabilities

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